NYU FRE 6103 – Valuation for Financial Engineers Name: _____
Midterm Exam – Fall 2025A – Professor Shimko

Email: _____@nyu.edu

Instructions: There is a two-page standard cheat sheet being distributed with this exam. You are allowed to use the cheat sheet and a calculator, but no use of telephones or other communication is allowed. Use the Blue Books provided to show your work, but put your final answers neatly on this sheet for grading. Blue Books will be checked only if necessary, but they should be turned in as well. Answer essay questions succinctly in the space provided. Time for exam: Two hours. There are 7 questions and 100 points.

1. (15 points) The triangular series is given by {0,1,3,6,10,15,21,…}. Compute the present value of cash flows defined by the triangular series, assuming the first payment is in one period, and the discount rate is "r" per period. Derive the MBE/GVE in your blue book and copy the result neatly here:

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2. (15 points) Imagine a one-period financial market in Lintner's CAPM, with only two securities and the parameters shown below. Assume that the risk-free interest rate is zero. Please complete the following tables. Hint: What is the cost of variance risk? _____

| Asset | Mean | Covariance | | Value | Market wt (%) | Exp Return (%) | Std Dev (%) |
|---|---|---|---|---|---|---|---|
| 1 | 50 | 100 | 20 | $V_1$=44 | | | |
| 2 | 60 | 20 | 144 | $V_2$= | | | |
| Mkt | | XXXXXX | | | | | |

3. (15 points) Compute the Cholesky decomposition of the covariance matrix $\begin{pmatrix} 100 & 60 \\ 60 & 360 \end{pmatrix}$

| | | |
|---|---|---|
| | | Let C represent the Cholesky decomposition matrix. What formula would you use to convert correlated normal random numbers (represented by an nx2 matrix "W") into uncorrelated random numbers "U"? |
| | | Ans: |

4. (10 points) In the simulation output below, fill in the blanks.

| | | |
|---|---|---|
| Sample size: | 500 | |
| Sample mean: | 135.20 | |
| Sample standard deviation: | | _____ |
| Sample standard error: | | _____ |
| Maxerr: | 3.14 | |

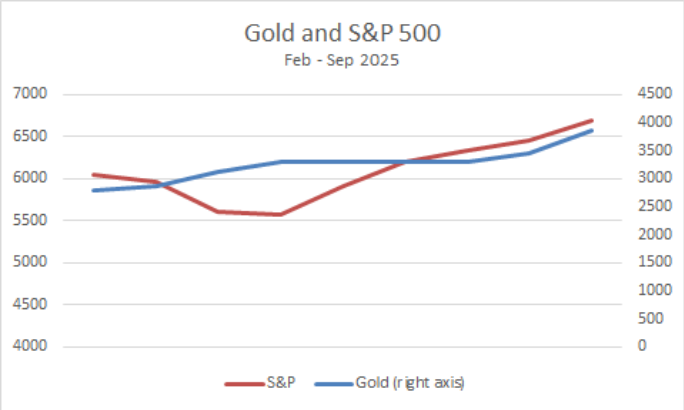
How many samples would be required to get maxerr down to 1.00? _____

5. (15 points)

An analyst wants to explore the relationship between gold prices and the stock market in 2025, and has produced the graph on the right.

He argues that the correlation between Gold prices and the S&P levels is 57.8%, suggesting that both are responding positively to the current economic environment.

Can you see any weak points in the analysis, or suggest any alternative explanations?




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6. (15 points) Let $S_t$ represent the price of corn at time t, $\mu$ represent the long run mean price, $\kappa$ the speed of mean reversion, and $\sigma$ the standard deviation. Write down the stochastic process for corn prices when there is a continuous probability $\pi\, dt$ of a jump of size J.

_____

Show what you would add to the stochastic process if there were a 30% chance that J was exponentially distributed with probability parameter $\lambda$, and 70% chance it was a constant, at $5.00.

_____

7. (15 points) What is the meaning of "capital" in a financial context (i.e. recall that *capital* is not the same as cash). Why is the concept of capital important to banks, and how should it affect the way we should think about valuation and performance measurement?

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_____

__What is the name of the measure for return on capital? _____

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1. (15 points) The triangular series is given by {0,1,3,6,10,15,21,…}. Compute the present value of cash flows defined by the triangular series, assuming the first payment is in one period, and the discount rate is "r" per period. Derive the MBE/GVE in your blue book and copy the result neatly here:

____$V_0$ = PV{0,1,3,6,10,15,21,…}_____ $V_1$ = PV{1,3,6,10,15,21,28…}_____

____ $V_1 - V_0$ = PV{1,2,3,4,5,6,…} = $(1+r)/r^2$ _____

_____$rV_0 + 0 = (1+r)/r^2 + 0$_____

_____$V_0 = (1+r)/r^3$_____

2. (15 points) Imagine a one-period financial market in Lintner's CAPM, with only two securities and the parameters shown below. Assume that the risk-free interest rate is zero. Please complete the following tables. Hint: What is the cost of variance risk? __A = 0.05__

| Asset | Mean | Covariance | | Value | Market wt (%) | Exp Return (%) | Std Dev (%) |
|---|---|---|---|---|---|---|---|
| 1 | 50 | 100 | 20 | $V_1$=44 | 46% | 14% | 23% |
| 2 | 60 | 20 | 144 | $V_2$=51.8 | 54% | 16% | 23% |
| Mkt | 110 | Grand Sum | 284 | M=95.8 | 100% | 15% | 18% |

3. (10 points) Compute the Cholesky decomposition of the covariance matrix $\begin{pmatrix} 100 & 60 \\ 60 & 360 \end{pmatrix}$

| | | |
|---|---|---|
| 10 | 0 | Let C represent the Cholesky decomposition matrix. What formula would you use to convert correlated normal random numbers (represented by an nx2 matrix "W") into uncorrelated random numbers "U"? |
| 6 | 18 | Ans: $U = W(C^T)^{-1}$ |

4. (10 points) In the simulation output below, fill in the blanks.

| | | |
|---|---|---|
| Sample size: | 500 | |
| Sample mean: | 135.20 | |
| Sample standard deviation: | | ___35.11_____ |
| Sample standard error: | | __1.57_____ |
| Maxerr: | 3.14 | |

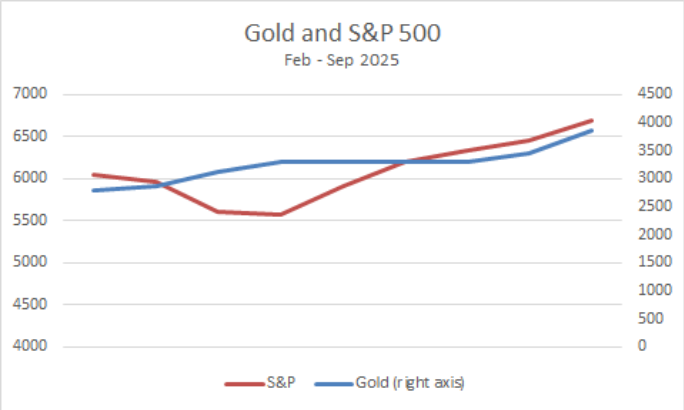
How many samples would be required to get maxerr down to 1.00? ___4930_____

5. (15 points)

An analyst wants to explore the relationship between gold prices and the stock market in 2025, and has produced the graph on the right.

He argues that the correlation between Gold prices and the S&P levels is 57.8%, suggesting that both are responding positively to the current economic environment.

Can you see any weak points in the analysis, or suggest any alternative explanation?




a. __ Observations are not i.i.d, since price levels are highly correlated
b. __ This is not consistent with the correct use of mathematical (Pearson's) correlation
c. __ Changes in prices are closer to i.i.d. when modeled
d. __ The correlation between price changes to be negative
e. __ It seems gold and stock are both rising, but with negatively correlated price changes.

6. (15 points) Let $S_t$ represent the price of corn at time t, $\mu$ represent the long run mean price, $\kappa$ the speed of mean reversion, and $\sigma$ the standard deviations. Write down the stochastic process for corn prices when there is a continuous probability $\pi\, dt$ of a jump of size J.

_____$S_{t+1} = S_t + \kappa(\pi - S_t) + \sigma z_{t+1} + I_J J$____($I_J$ or some kind of indicator)_____

Show what you would add to the stochastic process if there were a 30% chance that J was exponential with probability parameter $\lambda$, and 70% chance it was a constant, at $5.00.

_____ $I_J J$ = if(rand()<=0.3,-ln(rand()/$\lambda$,5)_____

7. (15 points) What is the meaning of "capital" in a financial context (i.e. recall that *capital* is not the same as cash). Why is the concept of capital important to banks, and how should it affect the way we should think about valuation and performance measurement?

__"Capital" in a banking setting refers to the maximum amount of risk a bank may take.

__Capital is a good denominator to use when measuring excess returns over the riskless rate, since it attempts

__to estimate the maximum loss.

__If capital is measured in dollars, all trade types can be evaluated in the same framework.

__The associated performance ratio --- the Sharpe Ratio or Dollarized Sharpe Ratio.

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